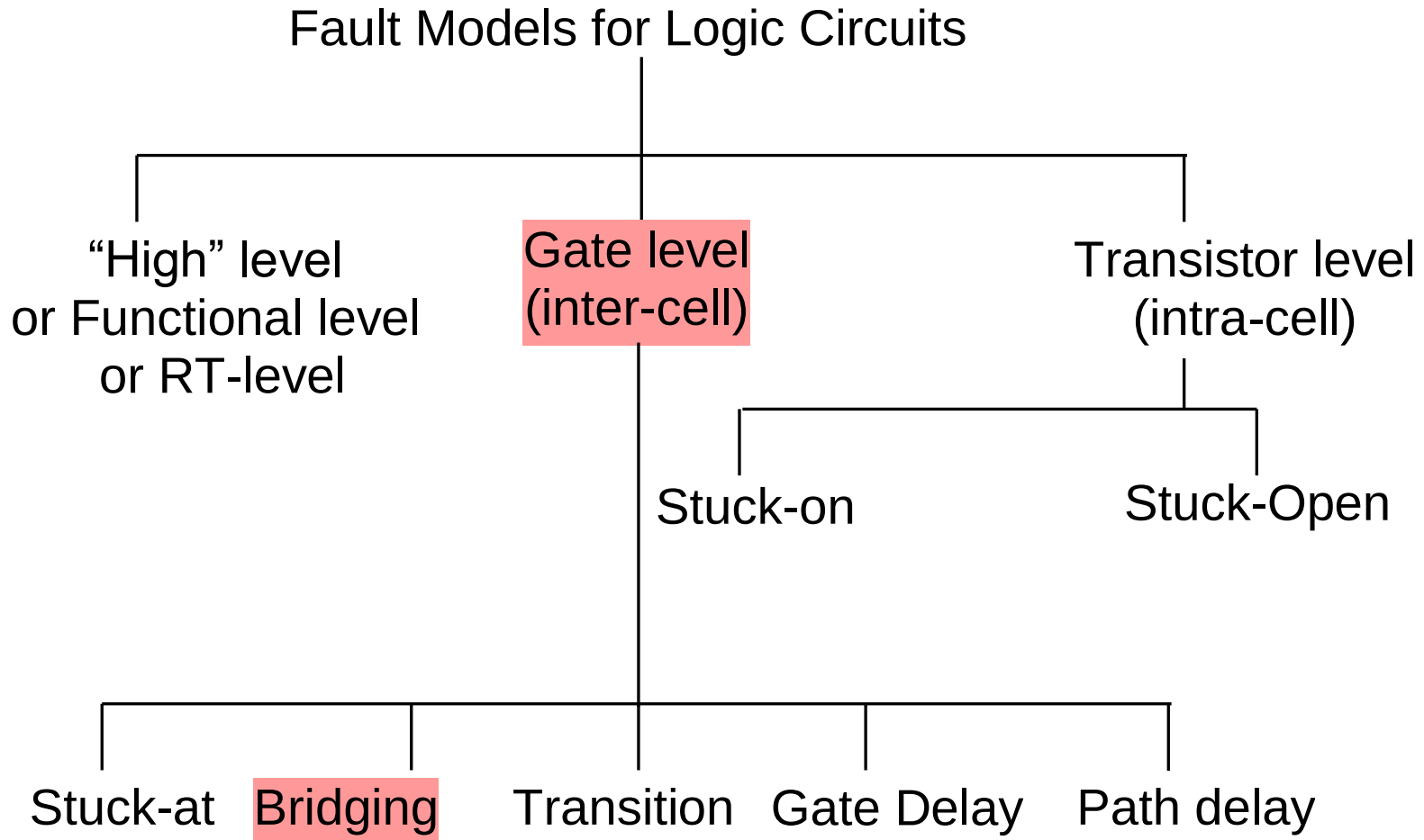


# Fault Modeling

- Introduction
- **Fault Models**
  - ◆ Stuck-at fault (1961)
  - ◆ **Bridging fault (1973)**
  - ◆ Delay fault (1974)
  - ◆ Transistor level fault
- Fault Detection
- Fault Coverage
- Conclusion



# Classification of Fault Models

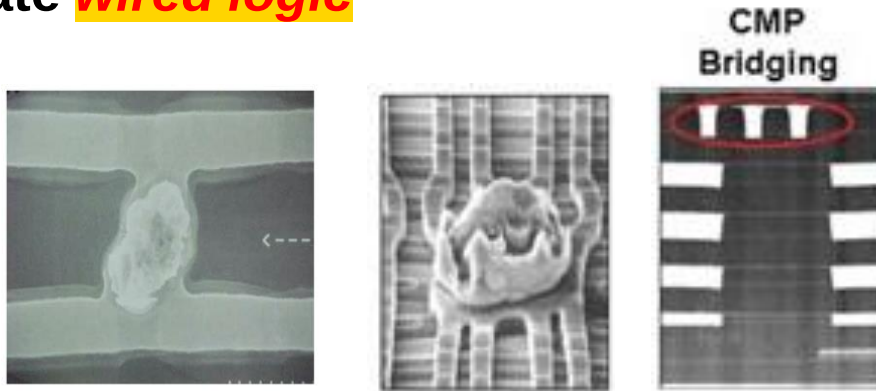


# Bridging Faults (BF)

[Williams 73][Friedman 74]

- Definition:

- ◆ Two (or more) distinct logic signals unintended shorted together and create **wired logic**



CMP= Chemical Mechanical Polishing  
Picture source: Mentor graphics

- Q: How many **two-way BF** in a circuit of  $n$  signals?
  - ◆ Arbitrary choose two signals:  $C^n_2 = O(n^2)$ . too many!
  - ◆ Need to identify pairs of neighbor signals **from layout**
    - \* **Fault extraction**

**Number of Bridging Faults is  $O(n^2)$**

# Bridging Fault Models

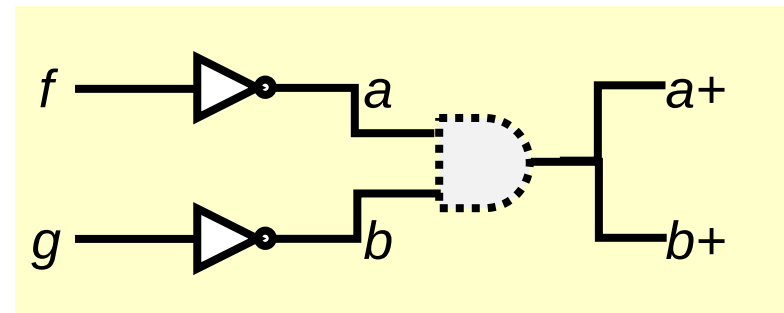
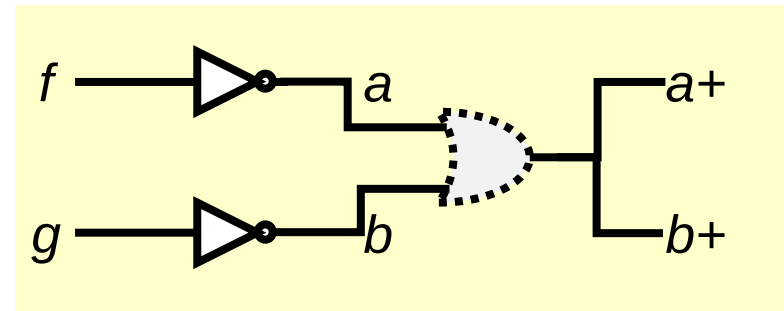
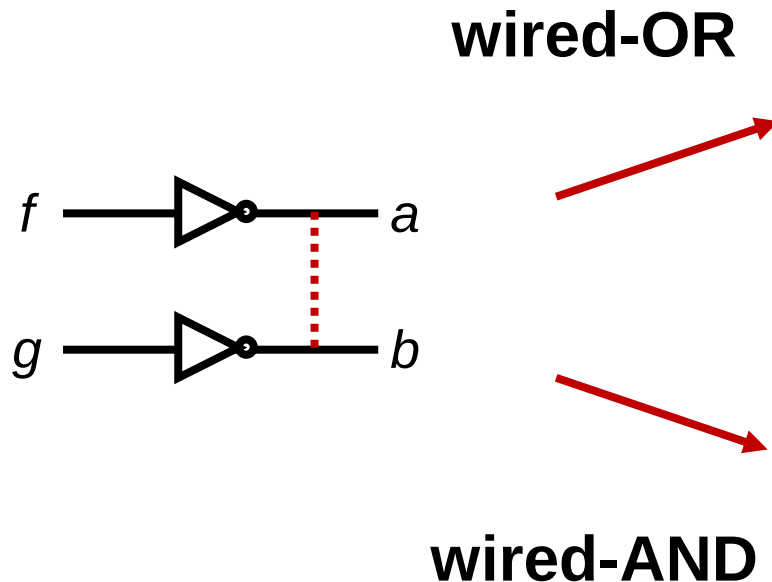
- Popular models for CMOS

- Wired-OR (1-dominant)

- Wired-AND (0-dominant)

- A-dominant

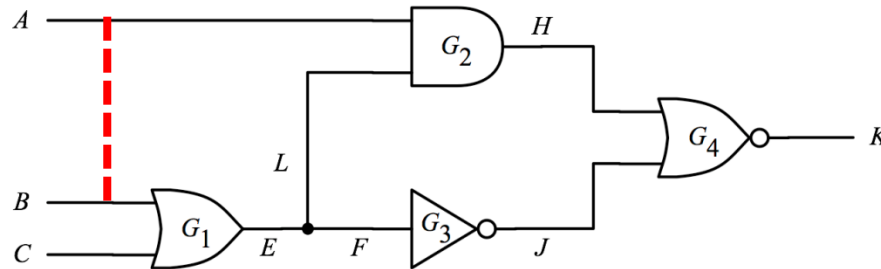
| original<br>a b | Wired-OR<br>a <sup>+</sup> b <sup>+</sup> | Wired-AND<br>a <sup>+</sup> b <sup>+</sup> | a-dominant<br>a <sup>+</sup> b <sup>+</sup> |
|-----------------|-------------------------------------------|--------------------------------------------|---------------------------------------------|
| 0 0             | 0 0                                       | 0 0                                        | 0 0                                         |
| 0 1             | <u>1</u> 1                                | 0 <u>0</u>                                 | 0 <u>0</u>                                  |
| 1 0             | 1 <u>1</u>                                | <u>0</u> 0                                 | 1 <u>1</u>                                  |
| 1 1             | 1 1                                       | 1 1                                        | 1 1                                         |



Wired Logic Is Imaginary, not Real

# Different Models Need Different Patterns

- Consider (A, B) bridging fault
  - Wired-AND, Wired-OR detected by **011**, or **010**
  - A-dominant model detected only by **010**

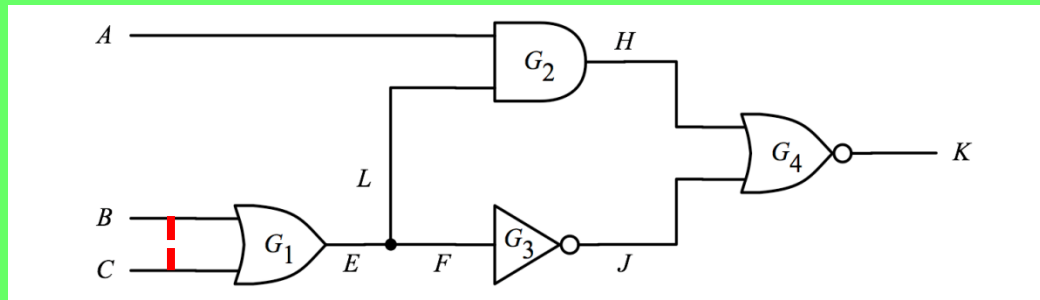


| Inputs<br>A B C | Fault-free<br>Output | Wired-<br>OR | Wired-<br>AND | A-<br>dominant |
|-----------------|----------------------|--------------|---------------|----------------|
| 0 0 0           | 0                    | 0            | 0             | 0              |
| 0 0 1           | 1                    | 1            | 1             | 1              |
| 0 1 0           | 1                    | <u>0</u>     | <u>0</u>      | <u>0</u>       |
| 0 1 1           | 1                    | <u>0</u>     | <u>0</u>      | 1              |
| 1 0 0           | 0                    | 0            | 0             | 0              |
| 1 0 1           | 0                    | 0            | 0             | 0              |
| 1 1 0           | 0                    | 0            | 0             | 0              |
| 1 1 1           | 0                    | 0            | 0             | 0              |

# Quiz

Q1: Fill in table with (B,C) bridging fault.

Q2: Which test pattern detects wired-OR fault?



| Inputs<br>A B C | Fault-free<br>Output | Wired-<br>OR | Wired-<br>AND | B-<br>dominant |
|-----------------|----------------------|--------------|---------------|----------------|
| 0 0 0           | 0                    | 0            | 0             | 0              |
| 0 0 1           | 1                    |              |               |                |
| 0 1 0           | 1                    |              |               |                |
| 0 1 1           | 1                    |              |               |                |
| 1 0 0           | 0                    |              |               |                |
| 1 0 1           | 0                    |              |               |                |
| 1 1 0           | 0                    |              |               |                |
| 1 1 1           | 0                    |              |               |                |

# SSF Test Sets Not Good Enough

- How effective is SSF test sets for bridging faults? [Millman 88]
  - ♦ 74LS181 ALU, 100% fault coverage SSF test sets
    - \* Total **7,981** testable bridging faults

| Test set       | Bry2 | Bry6 | Goel | Hugh | Krish | McC4 | Micz2 |
|----------------|------|------|------|------|-------|------|-------|
| Test Length    | 14   | 12   | 35   | 135  | 12    | 124  | 17    |
| # of missed BF | 111  | 138  | 15   | 2    | 171   | 13   | 85    |
| missed BF %    | 1.39 | 1.73 | 0.19 | 0.03 | 2.14  | 0.16 | 1.07  |

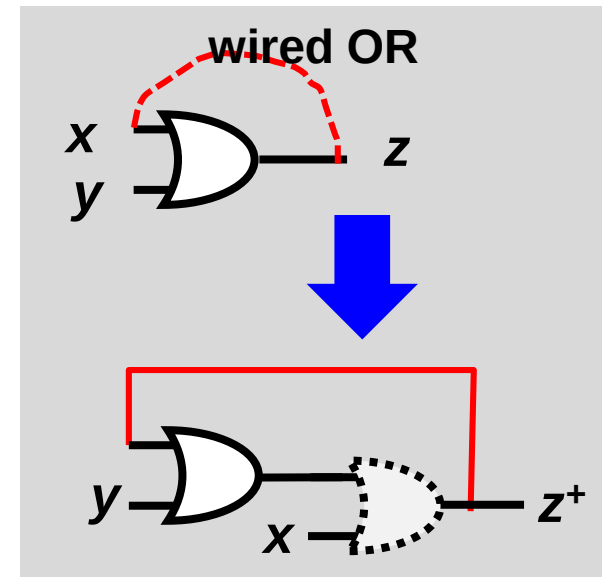
**Some BF not Detected by SSF Test Sets**

# Feedback Bridging Faults

- **Type1: creates memory**

- ◆ Detected by test sequence **01→00**

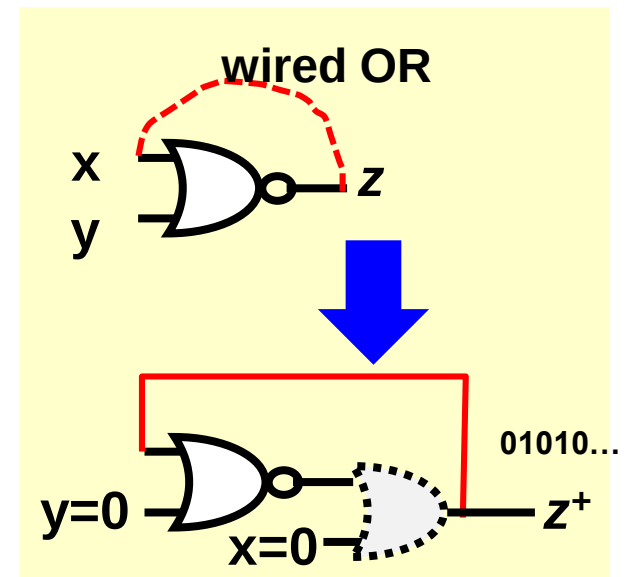
| Inputs<br>x y | Fault-free<br>Output | Faulty<br>Output |
|---------------|----------------------|------------------|
| 0 0           | 0                    | <u>unchanged</u> |
| 0 1           | 1                    | 1                |
| 1 0           | 1                    | 1                |
| 1 1           | 1                    | 1                |



- **Type2: creates oscillation**

- ◆ “hard” detected by **10**, or **11**
  - ◆ “potentially” detected by **00**

| Inputs<br>x y | Fault-free<br>Output | Faulty<br>Output |
|---------------|----------------------|------------------|
| 0 0           | 1                    | oscillation      |
| 0 1           | 0                    | 0                |
| 1 0           | 0                    | <u>1</u>         |
| 1 1           | 0                    | <u>1</u>         |



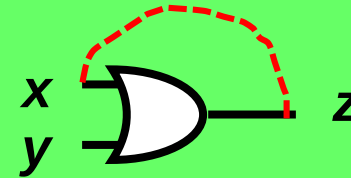


# Quiz

**Q1: Fill in table with a Wired-AND feedback BF ( $x, z$ )**

**A:**

| Inputs<br>$x\ y$ | Fault-free<br>Output | Faulty<br>Output |
|------------------|----------------------|------------------|
| 0 0              | 0                    |                  |
| 0 1              | 1                    |                  |
| 1 0              | 1                    |                  |
| 1 1              | 1                    |                  |

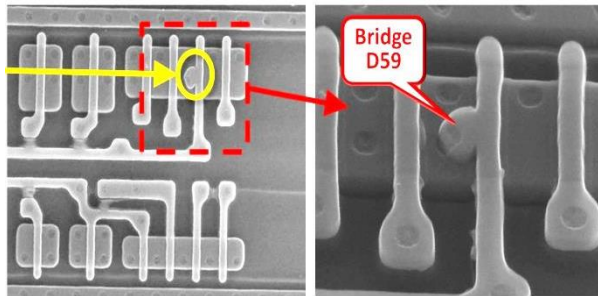


**Q2: Find a test to detect this feedback BF**

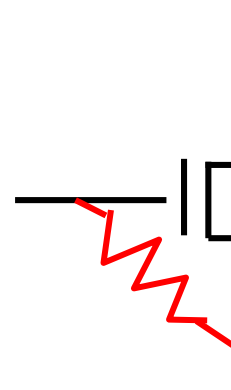
**A:**

# What are NOT Bridging Fault?

- ① BF does NOT consider shorts to **power** and **ground**
- ② BF does NOT consider **defect resistance value**
- ③ BF does NOT consider **Intra-cell** (intra-gate) defects
  - ◆ BF is **gate-level**, NOT **transistor-level** fault model



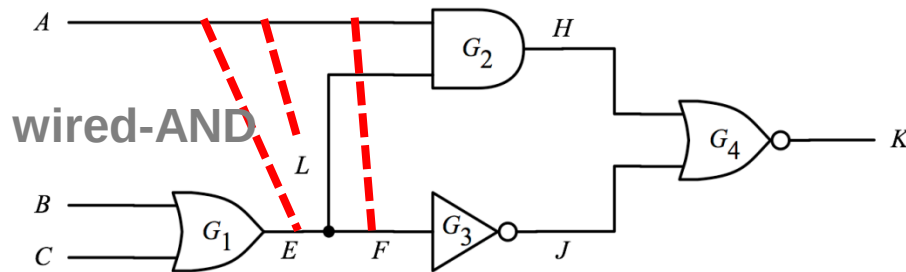
Intra-gate bridging defect  
[Mentor graphics]



- ④ BF does NOT distinguish between **fanout stem** and **branches**
  - ⑤ BF is **NOT transient** fault
- (4, 5 see next slides)

# Fanout Stem and Branches

- SSF on fanout stem and branches are different (see 3.2)
- BF on fanout stem and branches are the same
- Example: wired-AND model
  - ♦  $(A,E) = (A,L) = (A,F)$
  - ♦ Why? FFT



| Inputs<br>A B C | Fault-free<br>Output | BF<br>(A,E) | BF<br>(A,L) | BF<br>(A,F) |
|-----------------|----------------------|-------------|-------------|-------------|
| 0 0 0           | 0                    | 0           | 0           | 0           |
| 0 0 1           | 1                    | <u>0</u>    | <u>0</u>    | <u>0</u>    |
| 0 1 0           | 1                    | <u>0</u>    | <u>0</u>    | <u>0</u>    |
| 0 1 1           | 1                    | <u>0</u>    | <u>0</u>    | <u>0</u>    |
| 1 0 0           | 0                    | 0           | 0           | 0           |
| 1 0 1           | 0                    | 0           | 0           | 0           |
| 1 1 0           | 0                    | 0           | 0           | 0           |
| 1 1 1           | 0                    | 0           | 0           | 0           |

**BF on Stem and Branches Are Same**

# Permanent Faults vs. Transient Faults

- **Permanent faults:** faults always present

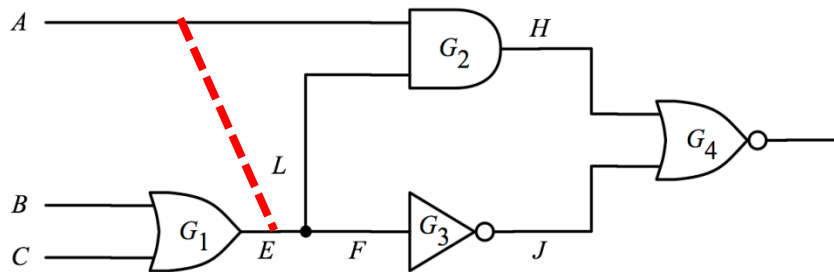
- ♦ caused by defects

- ♦ e.g. bridging faults caused by particle defects

- **Transient faults:** faults not always present

- ♦ induced by environmental (EMI) or internal (IR drop...) disturbance

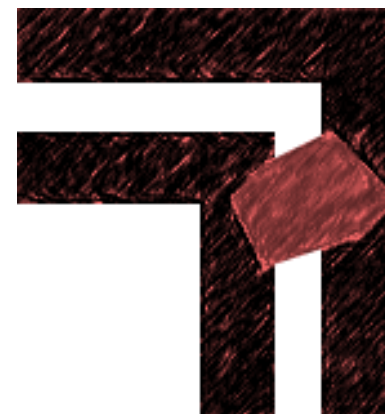
- ♦ e.g. crosstalk faults caused by coupling effect



| Inputs<br>A B C | Fault-free<br>Output | per-<br>manent | tran-<br>sient |
|-----------------|----------------------|----------------|----------------|
| 0 0 0           | 0                    | 0              | 0              |
| 0 0 1           | 1                    | <u>0</u>       | <u>0 or 1</u>  |
| 0 1 0           | 1                    | <u>0</u>       | <u>0 or 1</u>  |
| 0 1 1           | 1                    | <u>0</u>       | <u>0 or 1</u>  |
| 1 0 0           | 0                    | 0              | 0              |
| 1 0 1           | 0                    | 0              | 0              |
| 1 1 0           | 0                    | 0              | 0              |
| 1 1 1           | 0                    | 0              | 0              |

# Summary

- Bridging fault models for CMOS
  - ♦ **Wired-OR, wired-AND, A-dominant**
- Number of bridging faults is  $O(n^2)$ . too many!
  - ♦ Needs **fault extraction** from layout
- SSF test set may not good enough to detect all BF
  - ♦ **Feedback BF** may cause **memory** or **oscillation**
- BF model does NOT distinguish **fanout stem and branches**
- BF is **permanent fault** ; crosstalk fault is **transient fault**



# FFT

- BF does *NOT* distinguish fanout stem and branches
  - ♦  $(A,E) = (A,L) = (A,F)$
- FFT: BF is unlike SSF, why?
  - ♦  $E \text{ SA1} \neq F \text{ SA1} \neq L \text{ SA1}$  (see 3.2)

